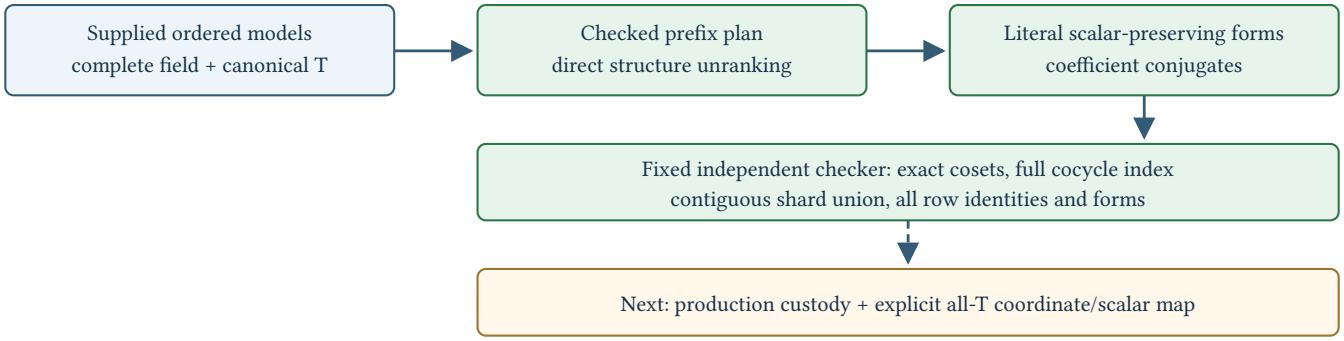
Streaming class × T × structure join

2026-11-02 | EXP-PEXT-bf8f96 | H-PEXT-c3a11d | TASK-20261102-d8d2b5

The arithmetic join now streams exact ordinal intervals over supplied ordered models, every canonical T coset and complete projective structures. Direct unranking skips structure prefixes; literal quartic coefficients and conjugates preserve scalars. No full prescribed-cohort join or cover/genus trial ran; the contract stays review_required/unapproved.



Software fixture	Classes	T blocks	Structures	Rows
F3 complete classes	8	14	1	14
F5 complete classes	12	24	1	24
F27 fixed char3 model	1	2	819	1,638

Direct ranges: immutable plan offsets are checked u64 sums/products; row=(class offset)+T_index*S+structure_index. Half-open bounds and row caps are checked, empty intervals explicit and callback errors immediate. A1/A2/B arithmetic unranking matches the old nested-loop order at every index in F3/F27/F125. Arbitrary17-row slices concatenate exactly; the F2197 final-index control checks suffix access only.

Scalar identity: for normalized N=(a,b;c,d), G(X,Z)=F_T(dX-bZ,-cX+aZ), so G(N(X,Z))=det(N)^4*F_T(X,Z). Original forms include the baseline infinity branch. Rows retain exact models/T, matrices, cocycles, literal forms and every coefficient conjugate; no monic normalization or genus/rank output.

Coverage and checks: independent polynomial arithmetic reconstructs points/doubles/cosets and exact fixture models, requires S distinct cocycles N^-1*sigma(N), consistent structure ordering and a disjoint contiguous shard union [0,total). Source-bound software-tests.txt records8Rust tests and25Python methods (6quotient,5structure,5class,4compact,5join), all PASS. Controls cover native shards, malformed identities/scalars/gaps, bounds, CLI and translation-involution equality on all fixed control points. The F27 fixture is one curve, not its full class cohort.

All-T distinction: P=T+2Q gives translation equivalence of involutions, checked at group level. Explicit quotient-coordinate/scalar transport remains to implement: T'=T+2Q induces A and requires structure reindexing HN -> HN*A^-1. Equality at the same fixed structure ordinal is not required. Verify rational function identities by clearing denominators modulo y^2=f(x), plus literal quartic identity with a nonzero square scale.

Next: production published-input/plan/file hash binding, streaming reducer/checkpoints and exact global structure/shard custody; the fixture JSON checker is bounded to10,000rows/structures. Preserve all five named cohorts, abstract n3/5/7 placements and original controls. Covering descent, genus and P3/H1 analysis follow. Full derivation, replay scope and archive prerequisites are in report.md; original snapshots and scientific graph/status remain unchanged.